

Stop-Loss: The Theory

gappy [+ Claude 4.8 + Codex 5.5]*

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Abstract

A hedge fund manager wants to hire a portfolio manager. The next-to-last question the portfolio manager asks the hedge fund manager is: “what is your drawdown policy?” Portfolio managers hate drawdowns, but hate stop-loss policies even more. In this note, I explain that stop-loss policies are necessary for the hedge fund manager, and are a good thing for the portfolio manager too. In the case of a “pure” stop-loss policy, i.e., one that allows the portfolio manager to liquidate the book, I show that the Sharpe Ratio is reduced at most by $(2 - \sqrt{3})/2 \approx 13\%$ in the strictest of the stop-loss policies, in exchange for peace of mind. I present a leading-order optimal vol adjustment policy in the presence of transaction costs. I show precisely how the stop-loss policy resembles a tail hedge on the portfolio. Cockroaches make an appearance.

1 If You Have to Read One Section, Make It This One

A toddler explores the world, and comes across a lit candle. Curious, he attempts to touch the flame with his finger. He feels the sudden sting of burning, and retracts his hand. This simple act, the withdrawal response, is the result of millions of years of evolution [5]. The receptor sends the signal to the spinal cord, which sends a signal back to the arm to retract it, and to the other limbs to simultaneously balance the body. The entire process happens within half a second and does not wait for the brain. It is a simple rule. It has saved millions of humans from life-altering injuries over the course of human history.

In his book *A Demon of Our Own Design*, Richard Bookstaber mentions the cockroach as an example of a resilient species [2].

What is remarkable about the cockroach is not only that it has survived for so long, but that it has done so with a singularly basic and seemingly suboptimal mechanism: Its defense mechanism is limited to moving away from slight puffs of air, puffs that might signal an approaching predator. This risk-management structure is extremely coarse; it ignores a wide set of information about the environment—visual and olfactory cues, for example—that one would think an optimal risk management system would take into account. The rule the cockroach obeys is so simple that it depends only on its giant fiber nervous system; it is a reaction that does not need to be filtered through its brain, but rather goes directly from the sensory hairs that detect the puff of air to the thoracic ganglia controlling its leg motion. This same pattern of behavior—using coarse decision rules that ignore valuable information—appears in other species with good track records of survivability. The crayfish, another old-timer in the evolutionary tree,

*I used Claude to flesh out the proofs. The only novel contribution in the paper is the extension of Grossman-Zhou to the case of quadratic costs, which is useful in practice but not very innovative.

uses a winner-take-all escape mechanism where a stimulus triggers a set of neurons, each dictating a pattern of action, and one variant of behavior then suppresses the circuits controlling the alternative actions. That is, although a number of different stimuli are received and processed, all but one of them are ignored.

Simple rules might not be optimal in some narrow sense, but they can be optimal in the broader sense that they maximize the user's chances of survival. This is the point Gigerenzer makes in his work on risk literacy: a good heuristic toolbox contains rules that are simple, transparent, and work in conditions of deep uncertainty [7]. And this brings us to stop-loss rules and the demonic world of finance. Stop-loss rules are perhaps the simplest of all risk management processes in finance, and among the least popular among portfolio managers. The two most important parameters a portfolio manager is given are:

1. a "size" budget. This can be expressed in terms of Gross Market Value of the portfolio, Long Market Value (usually paired with a requirement that the portfolio maintain a certain beta), volatility, or some other similar quantity.
2. a "loss" budget. This is almost always expressed as a dollar loss, which is itself a function of the size.

Portfolio managers like the freedom to make more money, and dislike the duty not to lose too much money in the process. The goal of this note is to analyze the properties of the "loss budget" rule. It does have some qualitative and quantitative merits.

Let us begin with the qualitative ones.

1. *First-principal protection.* The loss budget caps the damage any single book can do to the firm's capital pathwise, not merely in expectation or in distribution.
2. *Agent protection.* A pre-agreed mechanical rule also protects the portfolio manager: de-risking is mandated by policy rather than negotiated under duress, when judgment is at its worst.
3. *Agent incentives.* The portfolio manager's compensation is an option on performance; an explicit loss budget bounds the value of gambling for resurrection after losses.

This matters because the portfolio manager is not the only principal in the problem. A portfolio manager is an agent of the hedge fund, the hedge fund is an agent of its investors, and those investors are often agents of pensioners, citizens, or institutions whose capital they steward. The chain is long enough that incentives naturally drift. A stop-loss rule is a coarse way of keeping the delegated risk budget from becoming an open-ended promise. It is not a substitute for judgment, but it is a survival rule: simple, transparent, and hard to negotiate away after losses have already started.

The question is therefore not whether stop-loss rules are elegant. They are not. The question is whether there is a principled version of the rule, and what it costs. Grossman and Zhou answer exactly this question in a continuous-time model. The result is simple enough to state before doing any stochastic calculus.

1. Many PMs wait to cut their portfolio until they are just at the stop-loss limit. If you want to maximize portfolio growth, subject to a maximum loss, the optimal rule is proportional: reduce risk gradually as the loss limit approaches. In dollar-volatility units, the rule is

$$\text{dollar volatility} = \text{Sharpe ratio} \times \text{distance to the loss limit}$$

The closer the portfolio is to the loss boundary, the less risk it is allowed to take. At the max loss limit, risk is zero. Far from the boundary, risk rises in proportion to the cushion. This is the same intuition as portfolio insurance, but the multiplier is not a rule of thumb; it is the Kelly multiplier implied by the investment opportunity.

2. The loss-control constraint lowers performance, but not by as much as intuition suggests. In the pure frictionless model, the ordinary return Sharpe ratio under the optimal stop-loss rule is never below $\sqrt{3}/2$ of the unconstrained Sharpe ratio. Equivalently, even in the strictest stop-loss limit, the Sharpe penalty is at most

$$1 - \frac{\sqrt{3}}{2} \approx 13\%$$

This is the main practical surprise of the note. A hard drawdown rule sounds like a large tax on performance, but if the rule is implemented optimally, most of the risk-adjusted return survives.

3. The same rule has a tail-hedging interpretation. A stop-loss constraint can be viewed as a dynamically replicated protection strategy: hold a baseline risky position and add an overlay that becomes more short the risky asset as the portfolio draws down. A single vanilla put is not exactly the Grossman-Zhou policy, because the policy depends on the high-water mark and resets when a new high is reached. But a rolling put ladder gives the closest traded-option approximation.

2 The question Grossman and Zhou answer

A hedge fund manager wants to hire a portfolio manager. The next-to-last question the portfolio manager asks the hedge fund manager is: “what is your drawdown policy?” Portfolio managers hate drawdowns, but hate stop-loss policies even more. In this note, I explain that stop-loss policies are necessary for the hedge fund manager, and are a good thing for the portfolio manager too. The natural way to formulate “optimal growth with avoidance of a maximum loss” is not to prescribe an exposure rule directly. It is to choose the exposure rule that maximizes long-run growth subject to a hard drawdown constraint.

The original paper is Grossman and Zhou [8]. A clean extension and simplification is due to Cvitanic and Karatzas [4], and the semimartingale version is due to Cherny and Obloj [3]. Their problem is a drawdown-constrained version of the Merton/Kelly problem [10, 11].

Road map. Sections 3–5 set up the model, the drawdown floor, and the objective. Sections 6–8 state the optimal policy, long-run growth, volatility, and Sharpe ratios. Section 9 describes the stationary distribution of the drawdown state and of the policy. Section 10 gives the expected time to regain the high-water mark. Section 11 explains the option-protection interpretation. Section 12 gives the quadratic-cost extension, and Section 13 gives a compact numerical scale. The appendix contains the continuous-time derivations, first without trading costs and then with quadratic costs.

The unconstrained benchmark. First ignore drawdown control. A strategy is a predictable risky fraction w_t : at time t , the investor puts the dollar amount $\psi_t = w_t x_t$ into the risky asset and leaves the rest in cash. With one risky asset earning excess return

$$\mu dt + \sigma dW_t$$

discounted wealth satisfies

$$\frac{dx_t}{x_t} = w_t (\mu dt + \sigma dW_t) \quad (1)$$

If the investor uses a constant fraction w , Ito's formula gives the expected log-growth rate

$$\mu w - \frac{1}{2} \sigma^2 w^2 \quad (2)$$

Maximizing this quadratic function of w gives

$$w^K = \frac{\mu}{\sigma^2} \quad (3)$$

This is the Kelly strategy in this note: the constant risky fraction that maximizes expected log growth in the frictionless one-stock model. With Sharpe ratio $\theta = \mu/\sigma$, and with initial discounted wealth normalized to one, Kelly wealth therefore satisfies

$$\frac{dx_t^K}{x_t^K} = \theta^2 dt + \theta dW_t \quad (4)$$

Therefore

$$d \log x_t^K = \frac{1}{2} \theta^2 dt + \theta dW_t \quad (5)$$

and over a horizon T ,

$$\log x_T^K - \log x_0 = \frac{1}{2} \theta^2 T + \theta W_T \quad (6)$$

Thus the expected log growth per unit time is

$$g_K := \frac{1}{T} \mathbb{E}[\log x_T^K - \log x_0] = \frac{1}{2} \theta^2 \quad (7)$$

and the log-growth volatility per square-root time is

$$\sigma_{\log,K} := \sqrt{\frac{1}{T} \text{Var}(\log x_T^K - \log x_0)} = \theta \quad (8)$$

Equivalently, the volatility of the average growth rate $T^{-1}(\log x_T^K - \log x_0)$ is $\theta/\sqrt{T}$. The benchmark growth-to-log-volatility ratio is therefore

$$\boxed{\frac{g_K}{\sigma_{\log,K}} = \frac{\theta}{2} \quad \frac{g_K}{\theta \sigma_{\log,K}} = \frac{1}{2}} \quad (9)$$

The drawdown problem asks:

how much of this benchmark survives when wealth must stay above a moving floor.

3 Market model

The benchmark used the risky fraction w_t . For the drawdown problem it is more convenient to use dollar exposure, because the optimal rule will apply Kelly leverage to the cushion rather than to total wealth. There is a single traded asset. Per dollar invested it earns

$$\mu dt + \sigma dW_t \quad \mu > 0, \sigma > 0 \quad (10)$$

in excess of cash. Equivalently, x_t is discounted wealth: if the money-market account earns a deterministic rate r , then all wealths in this note should be read after discounting by that account. This is why uninvested capital earns zero in the notation. Define the Sharpe ratio

$$\theta := \frac{\mu}{\sigma} \quad (11)$$

Let ψ_t be the discounted dollar amount put at risk, so that $\sigma\psi_t$ is the dollar volatility of the position. Wealth then follows

$$dx_t = \psi_t (\mu dt + \sigma dW_t) \quad (12)$$

In terms of the volatility control $\nu_t := \sigma\psi_t$ this reads $dx_t = \nu_t(\theta dt + dW_t)$, so the only market parameter that matters is the Sharpe ratio θ .

4 Drawdown constraint

Let

$$M_t := \max_{0 \leq s \leq t} x_s \quad (13)$$

be the high-water mark of wealth. Fix a maximum tolerated drawdown $L \in (0, 1)$, and define the floor fraction

$$\alpha := 1 - L \quad (14)$$

The hard drawdown constraint is

$$x_t \geq \alpha M_t \quad \text{for all } t \geq 0 \quad (15)$$

Equivalently,

$$DD_t := 1 - \frac{x_t}{M_t} \leq L \quad (16)$$

For example, $L = 0.20$ means wealth is never allowed to fall more than 20% below its previous high.

The cushion above the drawdown floor is

$$C_t := x_t - \alpha M_t \quad (17)$$

The constraint says $C_t \geq 0$. At the drawdown floor, $C_t = 0$. At a new high-water mark, $x_t = M_t$, so $C_t = LM_t$.

5 Objective

The log-growth version of the Grossman-Zhou problem is

$$\sup_{\psi} \liminf_{T \rightarrow \infty} \frac{1}{T} \mathbb{E} [\log x_T] \quad (18)$$

subject to

$$x_t \geq \alpha M_t \quad \text{almost surely, for every } t \geq 0 \quad (19)$$

The supremum is over predictable trading rules that keep wealth positive and satisfy the drawdown constraint. This note does not dwell on the technical admissibility conditions; in the Black-Scholes case, the policy displayed below is admissible and attains the value. The important point is that the drawdown constraint is pathwise. It is not a penalty, not an expected shortfall constraint, and not a variance proxy. It is a hard rule on every realized path.

6 Optimal policy

Optimal Grossman-Zhou policy

At each time t , set the dollar exposure to the risky strategy to

$$\psi_t^* = \frac{\mu}{\sigma^2} (x_t - \alpha M_t) \quad (20)$$

Here, ψ_t^* is the optimal dollar exposure; μ is the strategy's expected excess return and σ is its volatility, measured over the same horizon; x_t is current portfolio capital; and M_t is its high-water mark. The maximum permitted drawdown is L (for example, $L = 20\%$), so $\alpha := 1 - L$ and αM_t is the dollar stop-loss floor. Thus $x_t - \alpha M_t$ is the remaining dollar cushion before the limit is reached: apply the Kelly multiplier μ/σ^2 to that cushion, and hold zero risk when the cushion is zero.

Equivalently, the optimal dollar volatility is

$$v_t^* := \sigma \psi_t^* = \theta (x_t - \alpha M_t) \quad (21)$$

The formula says: apply the Kelly multiplier μ/σ^2 , but apply it only to the cushion above the drawdown floor. This is why the rule looks like CPPI, but the multiplier is not a heuristic. It is the log-optimal Kelly multiplier inherited from the unconstrained auxiliary problem.

The risky fraction of current wealth is

$$w_t^* := \frac{\psi_t^*}{x_t} = \frac{\mu}{\sigma^2} \left(1 - \frac{\alpha M_t}{x_t} \right) \quad (22)$$

If

$$R_t := \frac{x_t}{M_t} \in [\alpha, 1] \quad (23)$$

then

$$w_t^* = \frac{\mu}{\sigma^2} \left(1 - \frac{\alpha}{R_t} \right) \quad (24)$$

The boundary behavior is the economic content:

$$R_t = \alpha \Rightarrow w_t^* = 0 \quad (25)$$

$$R_t = 1 \Rightarrow w_t^* = L \frac{\mu}{\sigma^2} \quad (26)$$

The investor goes flat at the floor and takes at most the fraction L of the unconstrained Kelly allocation at a high-water mark. The continuous-time derivation uses the Grossman-Zhou/Azema-Yor transform and is given in Appendix 15.

Writing current drawdown as $d_t := 1 - R_t$, the normalized policy is

$$\frac{w_t^*}{w^K} = \frac{L - d_t}{1 - d_t} \quad 0 \leq d_t \leq L \quad (27)$$

Figure 1 plots this rule for $L = 20\%$.

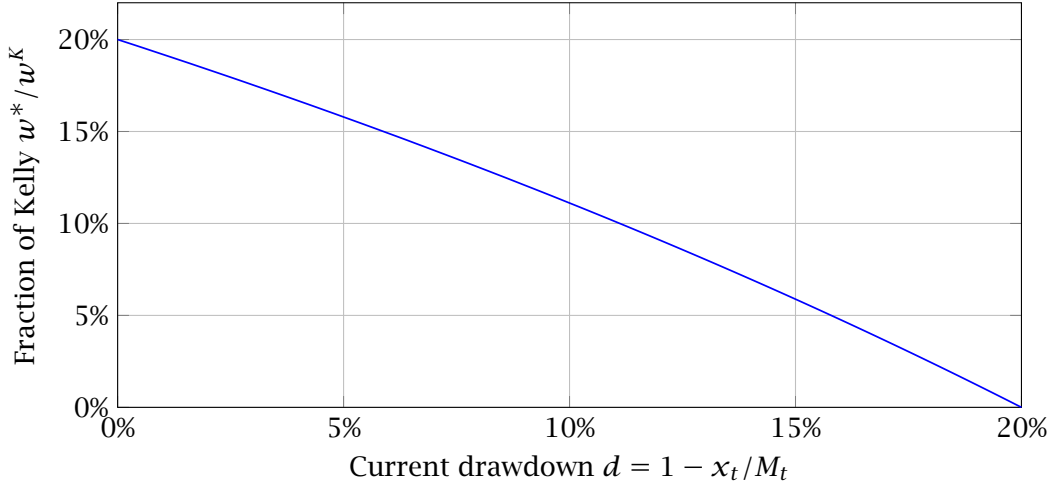


Figure 1: Optimal risky fraction as a function of current drawdown, normalized by the unconstrained Kelly fraction $w^K = \mu/\sigma^2$. The chart uses $L = 20\%$.

7 Long-run growth rate

The unconstrained benchmark from Section 1 has log-growth rate

$$g_{\text{Kelly}} = \frac{1}{2} \theta^2 \quad (28)$$

Under the drawdown constraint, the optimal long-run log-growth rate is

Growth cost of the drawdown constraint

$$g^* = L \frac{\theta^2}{2} = L \frac{\mu^2}{2\sigma^2} \quad (29)$$

Only the fraction L of the unconstrained Kelly growth premium survives the hard drawdown constraint.

Thus the drawdown constraint scales the growth rate by L . The two limiting cases are exactly what one should expect:

$$L = 0 \Rightarrow g^* = 0 \quad (30)$$

$$L = 1 \Rightarrow g^* = \frac{\mu^2}{2\sigma^2} \quad (31)$$

With no allowed drawdown, the investor stops trading. With no drawdown constraint, the investor recovers unconstrained Kelly.

8 Volatility under the optimal policy

Instantaneous log-volatility

The instantaneous log-volatility is

$$\sigma_{\log,t} = \sigma w_t^* = \theta \left(1 - \frac{\alpha}{R_t}\right) \quad (32)$$

Therefore

$$0 \leq \sigma_{\log,t} \leq L\theta \quad (33)$$

The maximum instantaneous log-volatility occurs at a high-water mark.

In terms of current drawdown $d_t = 1 - R_t$,

$$\frac{\sigma_{\log,t}}{\theta} = \frac{L - d_t}{1 - d_t} \quad 0 \leq d_t \leq L \quad (34)$$

Figure 2 plots the normalized instantaneous volatility for $L = 20\%$.

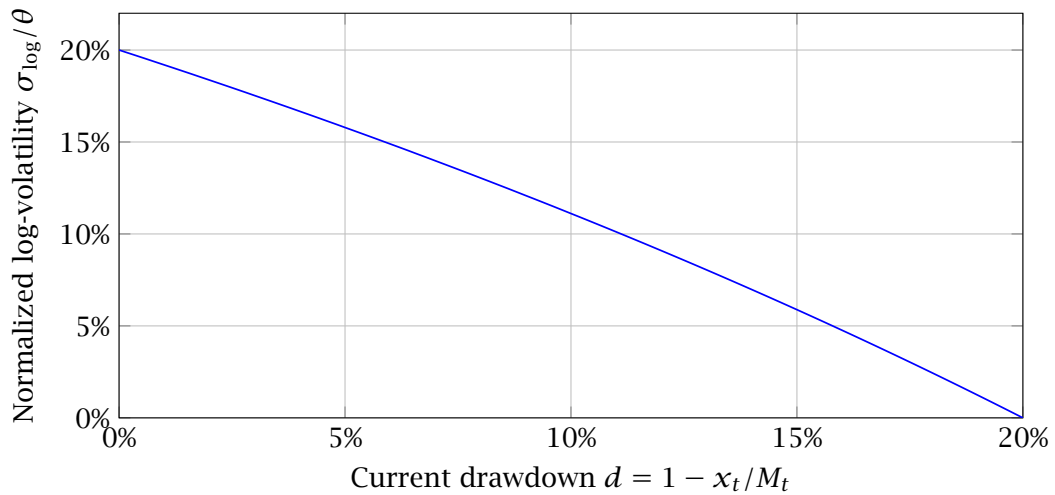


Figure 2: Instantaneous log-volatility under the optimal policy, normalized by the asset Sharpe ratio θ . The chart uses $L = 20\%$.

In the constant-coefficient one-stock model, an occupation-time average can be computed explicitly. Define

$$\bar{\sigma}_{\log}^2 := \lim_{T \rightarrow \infty} \frac{1}{T} \int_0^T \sigma_{\log,t}^2 dt \quad (35)$$

Then

$$\bar{\sigma}_{\log}^2 = \theta^2 \left[2 - L - \frac{2(1-L)}{L} \log \left(\frac{1}{1-L} \right) \right] \quad (36)$$

For small L ,

$$\bar{\sigma}_{\log} \sim \frac{\theta L}{\sqrt{3}} \quad (37)$$

The ordinary return Sharpe ratio is the arithmetic mean excess return divided by arithmetic-return volatility. As derived from the occupation calculation in Appendix 15.4, for $0 < L < 1$,

$$\text{SR}_{\text{ret}}(L) = \theta \frac{1 - \frac{1-L}{L} \log \left(\frac{1}{1-L} \right)}{\sqrt{2 - L - \frac{2(1-L)}{L} \log \left(\frac{1}{1-L} \right)}} \quad (38)$$

The endpoint limits are

$$\lim_{L \downarrow 0} \text{SR}_{\text{ret}}(L) = \frac{\sqrt{3}}{2} \theta \quad \text{SR}_{\text{ret}}(1) = \theta \quad (39)$$

Thus the ordinary return Sharpe is an increasing function of the allowed drawdown. A tight drawdown rule gives a normalized Sharpe near $\sqrt{3}/2$, while the unconstrained Kelly limit gives 1. This is a useful warning: drawdown control improves path protection and reduces absolute growth, but it also leaves the ordinary arithmetic-return Sharpe below the unconstrained Kelly Sharpe.

What is the log-return Sharpe?

For a dynamic wealth process

$$\frac{dx_t}{x_t} = b_t dt + s_t dW_t \quad (40)$$

write $\langle f \rangle := \lim_{T \rightarrow \infty} \frac{1}{T} \int_0^T f_t dt$ for the long-run occupation average. The ordinary return Sharpe and the log-return Sharpe are

$$\text{SR}_{\text{ret}} := \frac{\langle b \rangle}{\sqrt{\langle s^2 \rangle}} \quad (41)$$

$$\text{SR}_{\log} := \frac{\langle b - \frac{1}{2} s^2 \rangle}{\sqrt{\langle s^2 \rangle}} \quad (42)$$

Thus, for the same policy,

$$\text{SR}_{\log} = \text{SR}_{\text{ret}} - \frac{1}{2} \sqrt{\langle s^2 \rangle} \quad (43)$$

The denominator is the same. The numerator differs because the log-return Sharpe subtracts the Ito compounding drag.

For the Grossman-Zhou policy,

$$\text{SR}_{\log}(L) = \theta \frac{L/2}{\sqrt{2 - L - \frac{2(1-L)}{L} \log\left(\frac{1}{1-L}\right)}} \quad (44)$$

- *Pro: compounding-aware.* Its numerator is the actual long-run log-growth rate, so variance drag is treated as a real cost rather than as a separate risk diagnostic.
- *Pro: policy-aware.* For dynamic strategies, it uses occupation averages under the policy actually followed, so it sees time spent near the floor, at high exposure, or at low exposure.
- *Con: not leverage-invariant.* Scaling up a fixed strategy leaves SR_{ret} unchanged but lowers $\text{SR}_{\log}$ through the term $\frac{1}{2}\sqrt{\langle s^2 \rangle}$.
- *Con: not an unconstrained objective.* If leverage can be freely shrunk, maximizing $\text{SR}_{\log}$ can favor nearly flat portfolios. Growth $g = \langle b - \frac{1}{2}s^2 \rangle$, not the ratio, is the primitive Kelly objective.
- *Use: fixed-scale reporting.* It is useful when mandate, margin, drawdown, or risk-budget constraints already fix the scale of the strategy.
- *Use: comparing policy shapes.* It is useful for comparing drawdown control, volatility targeting, stop-out rules, and other state-dependent policies at a common risk budget.

For the Grossman-Zhou family, the log-return Sharpe has the opposite monotonicity from the ordinary return Sharpe: it falls from $\sqrt{3}\theta/2$ as $L \downarrow 0$ to $\theta/2$ at $L = 1$. There is no contradiction; the two Sharpe ratios answer different questions. Appendix 15.7 derives both formulas, plots the declining log-return Sharpe, and verifies the monotonicity by simulation.

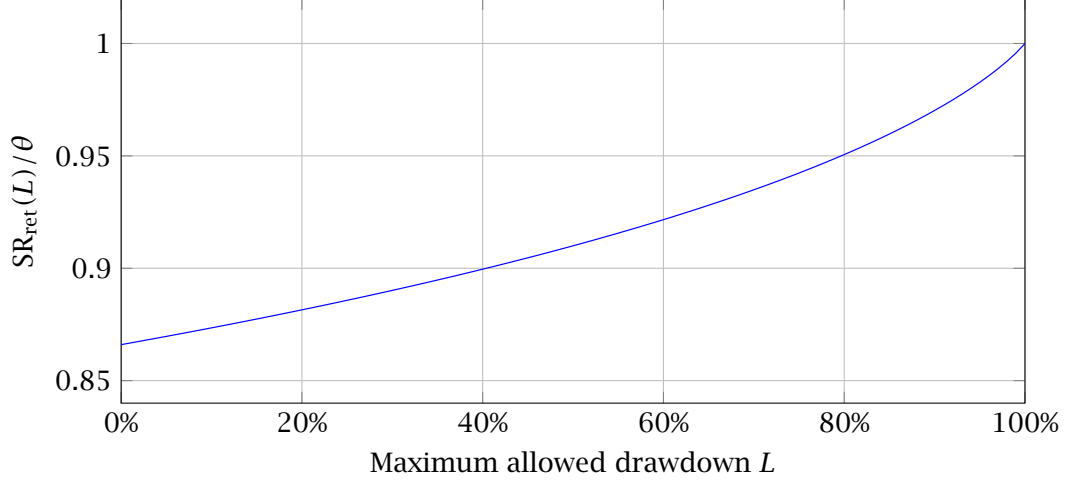


Figure 3: The normalized ordinary return Sharpe ratio $SR_{\text{ret}}(L)/\theta$ as a function of the maximum allowed drawdown L . It increases from $\sqrt{3}/2$ at $L \rightarrow 0$ to 1 at $L = 1$, where it matches the unconstrained Kelly return Sharpe.

9 Stationary Distribution Under the Optimal Policy

The optimal policy is easier to understand if we separate the state variable from the allocation induced by that state. Let

$$R_t := \frac{x_t}{M_t} \quad Q_t := \frac{R_t - \alpha}{L} \quad \alpha = 1 - L \quad (45)$$

Here R_t is current wealth as a fraction of the current high-water mark, and $Q_t \in (0, 1]$ rescales the allowed drawdown interval to the unit interval. The occupation-time calculation in [Appendix 15.4](#) shows that, under the Grossman-Zhou policy,

$$Q_t \text{ has stationary density } 1 \text{ on } (0, 1)$$

Thus

$$R_t = \alpha + LQ_t \quad D_t := 1 - R_t = L(1 - Q_t) \quad (46)$$

has a uniform stationary distribution on $(\alpha, 1)$ for R_t , or equivalently on $(0, L)$ for the current percentage drawdown D_t . This is the first useful check: the drawdown state itself is not positively skewed. It is uniform. The skewness enters through the nonlinear way the state is converted into risk.

The optimal risky fraction, measured in units of the unconstrained Kelly fraction, is

$$A_t := \frac{\text{Grossman-Zhou risky fraction}}{\text{Kelly risky fraction}} = 1 - \frac{\alpha}{R_t} = \frac{LQ_t}{\alpha + LQ_t} \quad (47)$$

The complementary term

$$B_t := 1 - A_t = \frac{\alpha}{R_t} \quad (48)$$

is the “floor pressure” or de-risking multiplier. It vanishes in the unconstrained Kelly limit ($\alpha = 0$) and takes values in $[\alpha, 1]$ otherwise. Since Q_t is uniform, for $0 < L < 1$

$$\mathbb{P}(A_t \leq a) = \frac{\alpha a}{L(1 - a)} \quad 0 \leq a \leq L \quad (49)$$

and therefore

$$f_A(a) = \frac{\alpha}{L(1 - a)^2} \quad 0 < a < L \quad (50)$$

The distribution of A_t is bounded above by L . It spends most of its time near the maximum allowed risk and has a left tail toward zero risk. Consequently the allocation multiplier A_t is left-skewed. The floor pressure $B_t = 1 - A_t$ is the same distribution reflected around $1/2$, so it has positive skewness. This is the natural stationary object if one wants to describe the stop-loss as an intermittent, positively skewed risk-control force: most of the time it is small, but occasionally it becomes large when the portfolio approaches the floor.

The first three raw moments of A_t are

$$m_1 = 1 - \frac{\alpha}{L} \log\left(\frac{1}{\alpha}\right) \quad (51)$$

$$m_2 = 2 - L - \frac{2\alpha}{L} \log\left(\frac{1}{\alpha}\right) \quad (52)$$

$$m_3 = \frac{1 + \frac{3}{2}\alpha - 3\alpha^2 + \frac{1}{2}\alpha^3 - 3\alpha \log(1/\alpha)}{L} \quad (53)$$

Equivalently, and more compactly,

$$m_k = \frac{1}{L} \int_{\alpha}^1 \left(1 - \frac{\alpha}{r}\right)^k dr \quad (54)$$

for $k = 1, 2, 3$. The allocation skewness is

$$\gamma_A(L) = \frac{m_3 - 3m_1m_2 + 2m_1^3}{(m_2 - m_1^2)^{3/2}} \quad (55)$$

and the floor-pressure skewness is

$$\gamma_B(L) = -\gamma_A(L) \quad (56)$$

L	$\mathbb{E}[A_t]$	$\text{sd}(A_t)$	γ_A	γ_B
5%	0.025	0.014	-0.036	0.036
10%	0.052	0.029	-0.073	0.073
20%	0.107	0.058	-0.155	0.155
40%	0.234	0.113	-0.356	0.356
60%	0.389	0.164	-0.649	0.649
80%	0.598	0.195	-1.191	1.191

Table 1: Stationary moments of the Grossman-Zhou allocation multiplier A_t and the floor-pressure multiplier $B_t = 1 - A_t$. The drawdown state D_t is uniform on $(0, L)$, so its skewness is zero. The positive skewness belongs to the floor-pressure term: usually small, occasionally large. The calculations are reproduced by `Code/stationary_distribution.py`.

The no-stop-loss policy is the limiting Kelly policy. There is no stationary wealth distribution, because wealth grows and diffuses forever. There is, however, a stationary drawdown ratio for the auxiliary Kelly wealth: $Q_t = V_t/\bar{V}_t$ is again uniform on $(0, 1)$, so the unconstrained Kelly drawdown $1 - Q_t$ is uniform on $(0, 1)$. What changes under Grossman-Zhou is not the shape of the drawdown occupation measure. The constraint compresses the admissible drawdown interval from $(0, 1)$ to $(0, L)$, and then maps that uniform state nonlinearly into the risky allocation. Without the stop-loss, $A_t \equiv 1$ and $B_t \equiv 0$. With the stop-loss, A_t is random and left-skewed, while the risk-control term B_t is random and positively skewed.

10 Expected Time to Recover

Fix the current high-water mark at M , and suppose the current percentage drawdown is $D \in (0, L)$. Thus current wealth and the cushion above the drawdown floor are

$$x_0 = (1 - D)M \quad C_0 := x_0 - \alpha M = (L - D)M. \quad (57)$$

Let

$$\tau_{\text{rec}} := \inf\{t \geq 0 : x_t = M\} \quad (58)$$

be the first time wealth regains the existing high-water mark. Before τ_{rec} , both the high-water mark and the floor αM are fixed. Under the optimal Grossman-Zhou policy,

$$\psi_t^* = \frac{\mu}{\sigma^2} C_t, \quad (59)$$

so the cushion satisfies

$$\frac{dC_t}{C_t} = \theta^2 dt + \theta dW_t, \quad (60)$$

$$d \log C_t = \frac{\theta^2}{2} dt + \theta dW_t. \quad (61)$$

Recovery occurs when the cushion rises from $(L - D)M$ to LM . The log-cushion must therefore cross the level

$$a(D, L) := \log \left(\frac{L}{L - D} \right) = -\log \left(1 - \frac{D}{L} \right). \quad (62)$$

A Brownian motion with drift $\theta^2/2$ has mean first-passage time equal to the barrier divided by the drift. Hence

Expected time to regain the high-water mark

$$\mathbb{E}_D[\tau_{\text{rec}}] = \frac{2}{\theta^2} \log \left(\frac{L}{L - D} \right) = -\frac{2}{\theta^2} \log \left(1 - \frac{D}{L} \right) \quad (63)$$

If θ is annualized, the answer is in years. The recovery time depends on D and L only through the fraction of the drawdown budget that has been used, D/L . For a shallow drawdown,

$$\mathbb{E}_D[\tau_{\text{rec}}] \sim \frac{2D}{\theta^2 L} \quad \text{as } D/L \downarrow 0. \quad (64)$$

It tends to zero as $D \downarrow 0$, and diverges as $D \uparrow L$, because the optimal risky exposure vanishes at the drawdown floor.

Let consider a concrete example. A long/short equity portfolio usually has a ratio $\$/\text{GMV}$ in the range 3%-5%. Consider a stop-loss policy that allows for a drawdown equal to a fraction κ of the portfolio's $\$/\text{vol}$. When the drawdown limit is reached, the volatility budget of the portfolio manager is reduced by a factor p , and the process repeats. Hence, the pairs (portfolio volatility, drawdown) take the form $(\sigma_0 p^n, \sigma_0 \kappa p^n)$. Allow, somewhat unrealistically, the process to repeat an arbitrary number of times. The maximum drawdown possible until volatility is set to zero is equal to $L = \sigma_0(\kappa + \kappa p + \dots) = \sigma_0 \kappa / (1 - p)$. For example, if $p = \kappa = 0.7$, then $L = (7/3)\sigma_0$. If we assume that $\sigma_0/\text{GMV}_0 = 0.04$, then $L \times \text{GMV}_0 = 2.333 \times 0.04 \times \text{GMV}_0$, i.e., $L = 9.3\%$. For $L = 8\%$, representative interior drawdowns give the following conditional expected recovery times.

Current drawdown D	$\theta = 0.25$	$\theta = 0.50$	$\theta = 0.75$	$\theta = 1.0$
1%	4.3	1.1	0.47	0.27
2%	9.2	2.3	1.0	0.58
3%	15	3.8	1.7	0.94
4%	22	5.6	2.5	1.4
5%	31	7.9	3.5	2.0

Table 2: Expected number of years to regain the existing high-water mark under the frictionless Grossman-Zhou policy, for maximum allowed drawdown $L = 8\%$ and annualized Sharpe ratio $\theta = \mu/\sigma$.

11 Grossman-Zhou as Tail Risk Protection

Closest vanilla approximation

If the hedge must be implemented as a constant risky position plus bought puts, the closest single-option approximation is a rolling moderately out-of-the-money put initiated at each high-water mark. Deep out-of-the-money puts react too late, and very long-expiry puts have too little gamma. The calibration should match the Grossman-Zhou delta and gamma in the drawdown region where protection matters most.

Start at a high-water mark. Write

$$x_0 = M_0 = M \quad F := \alpha M \quad C_0 := x_0 - F = LM \quad \beta := \frac{\mu}{\sigma^2} \quad (65)$$

While the high-water mark is fixed, the Grossman-Zhou risky dollar position is

$$\psi_t^* = \beta C_t \quad C_t := x_t - F \quad (66)$$

If the discounted risky asset satisfies $dS_t/S_t = \mu dt + \sigma dW_t$, the cushion follows

$$\frac{dC_t}{C_t} = \beta \frac{dS_t}{S_t} \quad (67)$$

Therefore, over one drawdown episode,

$$C_t = LM \left(\frac{S_t}{S_0} \right)^\beta \exp \left\{ -\frac{1}{2} \beta (\beta - 1) \sigma^2 t \right\} \quad (68)$$

and the exact Grossman-Zhou share position is

$$\Delta_t^{GZ} = \frac{\psi_t^*}{S_t} = \frac{\beta C_t}{S_t} \quad (69)$$

At the start of the drawdown, ignoring the deterministic Ito decay over short horizons, this is the power curve

$$\Delta^{GZ}(S) = \frac{\beta C_0}{S_0} \left(\frac{S}{S_0} \right)^{\beta-1} \quad (70)$$

This curve is put-like only when $\beta > 1$. In that case, as S falls, the desired share position falls. If $\beta = 1$, the Grossman-Zhou exposure is already a constant share position during the drawdown episode. If $\beta < 1$, a long put has the wrong local shape.

Now approximate the Grossman-Zhou position by a constant share position q plus n bought European puts, each with strike K and maturity T_p . In discounted Black-Scholes units, define

$$d_1(S, K, T_p) := \frac{\log(S/K) + \frac{1}{2}\sigma^2 T_p}{\sigma\sqrt{T_p}} \quad d_2(S, K, T_p) := d_1(S, K, T_p) - \sigma\sqrt{T_p} \quad (71)$$

The put price is

$$p(S, K, T_p) = K\Phi(-d_2(S, K, T_p)) - S\Phi(-d_1(S, K, T_p)) \quad (72)$$

so the approximate strategy has delta

$$\Delta^{app}(S) = q - n\Phi(-d_1(S, K, T_p)) \quad (73)$$

A useful local calibration is the following. Choose the drawdown level $d^* \in (0, L)$ where the approximation should be most accurate, and write

$$S^* := S_0(1 - d^*) \quad z^* := 1 - d^* \quad (74)$$

For a single put, center the option gamma near S^* . Since put gamma is largest when $d_1 = 0$, set

$$K = S^* \exp\left\{\frac{1}{2}\sigma^2 T_p\right\} \approx S_0(1 - d^*) \quad (75)$$

The approximation is usually best with d^* between $L/2$ and L . Choosing $d^* \approx L$ gives a deeper out-of-the-money put and protects the floor, but it misses the early Grossman-Zhou deleveraging. Choosing $d^* \approx L/2$ gives a more faithful approximation over the middle of the drawdown.

The Grossman-Zhou gamma at the target point is

$$\Gamma^{GZ}(S^*) := \frac{\partial \Delta^{GZ}}{\partial S}(S^*) = \frac{\beta(\beta-1)C_0}{S_0^2} (z^*)^{\beta-2} \quad (76)$$

while one put has gamma

$$\Gamma^p(S^*) = \frac{\varphi(d_1^*)}{S^* \sigma \sqrt{T_p}} \quad d_1^* := d_1(S^*, K, T_p) \quad (77)$$

Matching gamma gives the put notional

$$n = \frac{\Gamma^{GZ}(S^*)}{\Gamma^p(S^*)} = \frac{\beta(\beta-1)C_0}{S_0} \frac{\sigma \sqrt{T_p} (z^*)^{\beta-1}}{\varphi(d_1^*)} \quad (78)$$

With the strike choice above, $d_1^* = 0$. Finally choose the constant share position so the initial total delta equals the Grossman-Zhou delta at the high-water mark:

$$q = \frac{\beta C_0}{S_0} + n \Phi(-d_1(S_0, K, T_p)) \quad (79)$$

The put premium paid at the high-water mark is

$$\Pi_0 = n p(S_0, K, T_p) \quad (80)$$

and the remaining cash position is $M - qS_0 - \Pi_0$.

The maturity T_p is not a theoretical expiration date for Grossman-Zhou; it is a practical roll horizon. A very short maturity gives high gamma but creates roll risk. A very long maturity gives low gamma, so matching the Grossman-Zhou slope requires excessive notional. A reasonable rule is to choose T_p so that $\sigma \sqrt{T_p}$ is comparable to the drawdown band being hedged, or simply to use the expected time between high-water-mark resets. The position is then re-calibrated whenever a new high-water mark is reached.

The best implementation is therefore not one deep out-of-the-money put. It is a rolling put ladder, with strikes spread across the drawdown interval and notionals chosen to fit the Grossman-Zhou delta curve. The single-put formula above is the one-factor version of that approximation.

12 Quadratic transaction costs

The frictionless Grossman-Zhou rule jumps instantly to the cushion target. With quadratic transaction costs, that jump is no longer free. The current risky fraction becomes a state variable, because the investor must decide how fast to move from the current position toward the target.

Use the scale-free continuous-time convention

$$dw_t = u_t dt \quad (81)$$

and

$$\frac{dx_t}{x_t} = w_t(\mu dt + \sigma dW_t) - \frac{\kappa}{2} u_t^2 dt \quad (82)$$

where w_t is the risky fraction and u_t is the trading rate. The frictionless target is

$$y(r) := \frac{\mu}{\sigma^2} \left(1 - \frac{\alpha}{r}\right) \quad r := \frac{x}{M} \in [\alpha, 1] \quad (83)$$

For small κ , the leading continuous-time policy is partial adjustment toward this target:

Quadratic costs: tracking rule

$$dw_t = \lambda(R_t)(y(R_t) - w_t)dt \quad \lambda(r) = \frac{\sigma}{\sqrt{\kappa}} \sqrt{\frac{r}{r - \alpha}} \quad (84)$$

Large costs slow the adjustment; small costs make the policy track the cushion-Kelly target quickly.

The speed is also state-dependent: the policy tracks faster near the floor because being off target is more dangerous there. This is the natural economic correction to the frictionless rule.

The leading long-run growth loss is explicit:

Quadratic costs: leading growth loss

$$g_\kappa = g^* - \frac{\theta^4}{3\sigma} \alpha L^{3/2} \sqrt{\kappa} + o(\sqrt{\kappa}) \quad (85)$$

Quadratic costs reduce growth at order $\sqrt{\kappa}$, not order κ .

This square-root scaling is the same qualitative effect that appears in small-cost tracking and price-impact problems: as costs shrink, the optimal tracking band tightens, but the trading rate rises.

For the ordinary return Sharpe, write

$$A_1(L) := 1 - \frac{\alpha}{L} \log\left(\frac{1}{\alpha}\right) \quad (86)$$

$$A_2(L) := 2 - L - \frac{2\alpha}{L} \log\left(\frac{1}{\alpha}\right) \quad (87)$$

The leading small-cost approximation is

$$\widetilde{\text{SR}}_{\text{ret},\kappa}(L) = \frac{\theta^2 A_1(L) - [B_g(L) - \frac{1}{2}B_v(L)] \sqrt{\kappa}}{\sqrt{\theta^2 A_2(L) + B_v(L) \sqrt{\kappa}}} \quad (88)$$

where

$$B_g(L) := \frac{\theta^4}{3\sigma} \alpha L^{3/2} \quad (89)$$

$$B_v(L) := \frac{\theta^4}{7\sigma} \alpha L^{5/2} \quad (90)$$

This formula is a local small-cost approximation, not an exact solution of the two-state HJB. The growth coefficient B_g is the genuine leading-order term: the envelope argument protects the objective against the order- $\sqrt{\kappa}$ perturbation of the occupation measure. The variance correction $B_v \sqrt{\kappa}$ is not similarly protected: it keeps the tracking-error variance but ignores the same-order shift in the occupation measure of R_t , so the growth formula above is exact at order $\sqrt{\kappa}$ while the Sharpe approximation is heuristic at that order. The continuous-time derivation is in Appendix 15.5; the Sharpe assembly and the plotted curves are in Appendix 15.6.

13 Numerical example

Use the following annualized parameters:

$$\mu = 8\%, \quad \sigma = 20\%, \quad L = 20\% \quad (91)$$

Then

$$\theta = \frac{\mu}{\sigma} = 0.40 \quad \frac{\mu}{\sigma^2} = 2.00 \quad (92)$$

The policy invests zero at the floor and at most

$$L \frac{\mu}{\sigma^2} = 0.20 \times 2.00 = 40\% \quad (93)$$

of current wealth in the risky asset. The optimal growth rate is

$$g^* = 0.20 \times \frac{0.40^2}{2} = 1.60\% \quad (94)$$

The maximum instantaneous log-volatility is

$$L\theta = 8.00\% \quad (95)$$

and the long-run average log-volatility is

$$\bar{\sigma}_{\log} = 4.88\% \quad (96)$$

The table uses the closed-form formulas above. It is meant only to put the scale of the result in familiar annualized units.

Quantity	Formula	Value
Floor fraction	$\alpha = 1 - L$	80.00%
Unconstrained Kelly fraction	μ/σ^2	200.00%
Max constrained risky fraction	$L\mu/\sigma^2$	40.00%
Unconstrained Kelly growth	$\theta^2/2$	8.00%
Constrained growth	$L\theta^2/2$	1.60%
Max instantaneous log-volatility	$L\theta$	8.00%
Average log-volatility	$\bar{\sigma}_{\log}$	4.88%

Table 3: Numerical values for $\mu = 8\%$, $\sigma = 20\%$, and $L = 20\%$.

14 Takeaways

The Grossman-Zhou solution replaces a naive drawdown-reactive rule with an optimal-control rule. The policy is not “invest less after losing money” in an ad hoc sense. It is exactly Kelly applied to the cushion over the drawdown floor:

$$\psi_t^* = \frac{\mu}{\sigma^2} (x_t - \alpha M_t) \quad (97)$$

The cost of hard drawdown protection is transparent: if the maximum allowed drawdown is L , only the fraction L of the unconstrained Kelly growth premium remains. The benefit is also transparent: realized log-volatility is clipped near the floor, and the dimensionless growth-to-log-volatility ratio rises as the drawdown limit tightens.

Quadratic transaction costs change the nature of the problem. The investor can no longer jump to the cushion target; the current risky fraction becomes an additional state variable. The small-cost continuous-time approximation says that the optimal policy tracks the Grossman-Zhou target at speed of order $\kappa^{-1/2}$, and the leading growth loss is explicit:

$$g^* - g_\kappa \sim \frac{\theta^4}{3\sigma} \alpha L^{3/2} \sqrt{\kappa} \quad (98)$$

15 Appendix: Continuous-Time Derivations

This appendix gives the continuous-time derivations behind the formulas in the main text. The rigorous versions use the Azema–Yor transform and the drawdown-constrained portfolio correspondence in Grossman and Zhou, Cvitanic and Karatzas, and Cherny and Obloj [8, 4, 1, 3]. In this appendix, the market has one discounted risky asset

$$\frac{dS_t}{S_t} = \mu dt + \sigma dW_t \quad \theta := \frac{\mu}{\sigma} > 0 \quad (99)$$

and all wealth processes are positive and continuous.

15.1 The drawdown transform

Let V_t be a positive unconstrained wealth process and define its running maximum by

$$\bar{V}_t := \max_{0 \leq s \leq t} V_s \quad (100)$$

For the linear drawdown floor αM , with $\alpha = 1 - L$, the Grossman–Zhou transform is generated by

$$F(v) = v^L, \quad 0 < L \leq 1 \quad (101)$$

Define the transformed wealth

$$x_t = F(\bar{V}_t) - F'(\bar{V}_t)(\bar{V}_t - V_t) \quad (102)$$

$$= (1 - L)\bar{V}_t^L + L\bar{V}_t^{L-1}V_t \quad (103)$$

Since $V_t \leq \bar{V}_t$,

$$\alpha \bar{V}_t^L \leq x_t \leq \bar{V}_t^L \quad (104)$$

Moreover equality on the upper bound holds exactly when $V_t = \bar{V}_t$. Hence the running maximum of the transformed process is

$$M_t = \bar{V}_t^L \quad (105)$$

and the transformed process satisfies

$$x_t \geq \alpha M_t \quad (106)$$

Thus the transform maps an unconstrained wealth path into a path satisfying the drawdown floor.

The key identity is the semimartingale differential

$$dx_t = F'(\bar{V}_t)dV_t \quad (107)$$

Indeed, applying integration by parts to

$$x_t = F(\bar{V}_t) - F'(\bar{V}_t)(\bar{V}_t - V_t)$$

gives

$$dx_t = F'(\bar{V}_t)dV_t - F''(\bar{V}_t)(\bar{V}_t - V_t)d\bar{V}_t \quad (108)$$

The finite-variation process $\bar{V}$ can increase only on the set $\{t : V_t = \bar{V}_t\}$. On that set $\bar{V}_t - V_t = 0$, so the second term vanishes. Therefore, if V_t is generated by an unconstrained risky-dollar position ψ_t^V , then the transformed wealth x_t is generated by

$$\psi_t = F'(\bar{V}_t)\psi_t^V \quad (109)$$

15.2 The continuous-time optimal policy

In the auxiliary unconstrained problem, write the risky fraction as

$$w_t^V := \frac{\psi_t^V}{V_t} \quad (110)$$

The log-wealth drift is

$$\mu w_t^V - \frac{1}{2} \sigma^2 (w_t^V)^2 \quad (111)$$

which is strictly concave in w_t^V . Its pointwise maximizer is the continuous-time Kelly/Merton fraction

$$w^K = \frac{\mu}{\sigma^2} \quad (112)$$

Thus the auxiliary optimal risky-dollar position is

$$\psi_t^{V,*} = \frac{\mu}{\sigma^2} V_t \quad (113)$$

Using the transform,

$$\psi_t^* = F'(\bar{V}_t) \psi_t^{V,*} \quad (114)$$

$$= \frac{\mu}{\sigma^2} L \bar{V}_t^{L-1} V_t \quad (115)$$

But the cushion of the transformed wealth above its drawdown floor is

$$x_t - \alpha M_t = L \bar{V}_t^{L-1} V_t \quad (116)$$

Therefore

$$\psi_t^* = \frac{\mu}{\sigma^2} (x_t - \alpha M_t) \quad (117)$$

This is the Grossman-Zhou policy. The only nontrivial optimality step is the transform correspondence: admissible drawdown-constrained wealth processes can be represented through an unconstrained auxiliary wealth process, and log utility is monotone under that correspondence. Once that reduction is accepted, the constrained problem inherits the ordinary Kelly solution in the auxiliary state.

15.3 The long-run growth rate

Under auxiliary Kelly,

$$\frac{dV_t}{V_t} = \frac{\mu}{\sigma^2} \mu dt + \frac{\mu}{\sigma^2} \sigma dW_t \quad (118)$$

so

$$\log V_T = \log V_0 + \frac{\theta^2}{2} T + \theta W_T \quad (119)$$

Since $W_T/T \rightarrow 0$ almost surely,

$$\frac{1}{T} \log V_T \rightarrow \frac{\theta^2}{2} \quad \text{a.s.} \quad (120)$$

The running maximum has the same exponential growth rate. To see this, fix $\varepsilon > 0$. Almost surely, for all sufficiently large t ,

$$\log V_t \leq \log V_0 + \left(\frac{\theta^2}{2} + \varepsilon \right) t \quad (121)$$

The finitely many earlier values only contribute $O(1)$ to the maximum, so

$$\limsup_{T \rightarrow \infty} \frac{1}{T} \log \bar{V}_T \leq \frac{\theta^2}{2} + \varepsilon \quad (122)$$

The reverse inequality follows from $\bar{V}_T \geq V_T$. Letting $\varepsilon \downarrow 0$,

$$\frac{1}{T} \log \bar{V}_T \rightarrow \frac{\theta^2}{2} \quad \text{a.s.} \quad (123)$$

For $0 < L < 1$, the bounds

$$\alpha \bar{V}_T^L \leq x_T \leq \bar{V}_T^L \quad (124)$$

give

$$\frac{1}{T} \log x_T \rightarrow L \frac{\theta^2}{2} \quad (125)$$

The endpoint $L = 1$ is the unconstrained Kelly case and gives the same formula by direct substitution.

The objective is stated in expectation, and the same sandwich applies there. Under auxiliary Kelly, $\mathbb{E}[\log V_T] = \log V_0 + \frac{\theta^2}{2} T$ exactly, while

$$\mathbb{E}[\log \bar{V}_T] \leq \log V_0 + \frac{\theta^2}{2} T + \theta \mathbb{E} \left[\max_{0 \leq s \leq T} W_s \right] = \log V_0 + \frac{\theta^2}{2} T + \theta \sqrt{\frac{2T}{\pi}} \quad (126)$$

because $\max_{s \leq T} \{ \frac{\theta^2}{2} s + \theta W_s \} \leq \frac{\theta^2}{2} T + \theta \max_{s \leq T} W_s$. The bounds $\alpha \bar{V}_T^L \leq x_T \leq \bar{V}_T^L$ then give $\frac{1}{T} \mathbb{E}[\log x_T] \rightarrow L \theta^2/2$ as well. Hence

$$g^* = L \frac{\theta^2}{2} \quad (127)$$

15.4 The occupation-time calculation

Let

$$R_t := \frac{x_t}{M_t} \quad Q_t := \frac{R_t - \alpha}{L} \quad (128)$$

Using the transform and $M_t = \bar{V}_t^L$,

$$R_t = \alpha + L \frac{V_t}{\bar{V}_t} \quad Q_t = \frac{V_t}{\bar{V}_t} \quad (129)$$

Thus $Q_t \in (0, 1]$. Set $U_t := \log Q_t = \log V_t - \log \bar{V}_t \leq 0$. Under auxiliary Kelly,

$$d \log V_t = \frac{\theta^2}{2} dt + \theta dW_t \quad (130)$$

Let

$$K_t := \log \bar{V}_t - \log \bar{V}_0 \quad (131)$$

Then K is nondecreasing and increases only when $U_t = 0$. Hence

$$dU_t = \frac{\theta^2}{2} dt + \theta dW_t - dK_t \quad dK_t \text{ is carried by } \{U_t = 0\} \quad (132)$$

This is Brownian motion with positive drift, reflected downward at the upper boundary 0 [9].

The stationary density p on $(-\infty, 0]$ is obtained from the zero-flux equation for the reflected diffusion:

$$\left(\frac{\theta^2}{2}\right) p(u) - \frac{1}{2} \theta^2 p'(u) = 0 \quad (133)$$

Thus $p'(u) = p(u)$, and normalization gives

$$p(u) = e^u, \quad u \leq 0 \quad (134)$$

Consequently $Q_t = e^{U_t}$ has uniform long-run occupation density on $(0, 1)$, because $e^u du = dq$.

The log-volatility coefficient of the drawdown-controlled wealth is

$$\sigma_{\log, t} = \theta \left(1 - \frac{\alpha}{R_t}\right) = \theta \frac{LQ_t}{\alpha + LQ_t} \quad (135)$$

Therefore its long-run variance rate is

$$\bar{\sigma}_{\log}^2 = \theta^2 \int_0^1 \left(\frac{Lq}{\alpha + Lq}\right)^2 dq \quad (136)$$

With $r = \alpha + Lq$, $dq = dr/L$, and $r \in [\alpha, 1]$,

$$\int_0^1 \left(\frac{Lq}{\alpha + Lq}\right)^2 dq = \frac{1}{L} \int_{\alpha}^1 \left(1 - \frac{\alpha}{r}\right)^2 dr \quad (137)$$

$$= 2 - L - \frac{2\alpha}{L} \log\left(\frac{1}{\alpha}\right) \quad (138)$$

Substituting $\alpha = 1 - L$ gives

$$\bar{\sigma}_{\log}^2 = \theta^2 \left[2 - L - \frac{2(1-L)}{L} \log\left(\frac{1}{1-L}\right)\right] \quad (139)$$

The same density gives the average arithmetic excess return. The local arithmetic drift under the optimal policy is $\theta^2 Lq/(\alpha + Lq)$, and the companion first-moment integral is

$$\int_0^1 \frac{Lq}{\alpha + Lq} dq = \frac{1}{L} \int_{\alpha}^1 \left(1 - \frac{\alpha}{r}\right) dr = 1 - \frac{\alpha}{L} \log\left(\frac{1}{\alpha}\right) \quad (140)$$

Dividing the average drift by $\bar{\sigma}_{\log}$ yields the main-text formula for $\text{SR}_{\text{ret}}(L)$; the full assembly, together with the log-return Sharpe, is in Appendix 15.7.

The same occupation calculation verifies the growth rate directly. If

$$a(q) := \frac{Lq}{\alpha + Lq} \quad (141)$$

then the instantaneous log-growth rate is

$$\theta^2 a(q) - \frac{1}{2} \theta^2 a(q)^2 \quad (142)$$

Using the uniform occupation density of q ,

$$\int_0^1 \left[\theta^2 a(q) - \frac{1}{2} \theta^2 a(q)^2 \right] dq = L \frac{\theta^2}{2} \quad (143)$$

which is consistent with the pathwise growth calculation above.

15.5 Continuous-time quadratic costs

Use a scale-free quadratic cost, charged as a fraction of current wealth. Let

$$w_t := \frac{\psi_t}{x_t} \quad (144)$$

be the risky fraction and let u_t be its trading rate:

$$dw_t = u_t dt \quad (145)$$

For a cost parameter $\kappa > 0$, wealth evolves as

$$\frac{dx_t}{x_t} = w_t(\mu dt + \sigma dW_t) - \frac{\kappa}{2} u_t^2 dt \quad (146)$$

This convention preserves scale invariance. If costs are instead charged on dollar turnover $\dot{\psi}_t$, the problem is no longer homogeneous in wealth.

The frictionless drawdown-constrained risky fraction is

$$y(r) := w^0(r) = \frac{\mu}{\sigma^2} \left(1 - \frac{\alpha}{r} \right) \quad r := \frac{x}{M} \in [\alpha, 1] \quad (147)$$

Quadratic costs prevent the investor from jumping instantly to this target. The state must include both r and the current risky fraction w . Away from the high-water boundary,

$$dR_t = R_t \left(\mu w_t - \frac{\kappa}{2} u_t^2 \right) dt + R_t \sigma w_t dW_t \quad (148)$$

The log-growth reward rate is

$$\mu w_t - \frac{1}{2} \sigma^2 w_t^2 - \frac{\kappa}{2} u_t^2 \quad (149)$$

Thus the ergodic HJB for the average growth rate g_κ has the form

$$g_\kappa = \sup_u \left\{ \mu w - \frac{1}{2} \sigma^2 w^2 - \frac{\kappa}{2} u^2 \right. \quad (150)$$

$$\left. + r \left(\mu w - \frac{\kappa}{2} u^2 \right) H_r + u H_w + \frac{1}{2} \sigma^2 w^2 r^2 H_{rr} \right\} \quad r \in (\alpha, 1) \quad (151)$$

The upper boundary $r = 1$ is the reflecting high-water boundary, with Neumann condition $H_r(1^-) = 0$. The lower boundary is the state constraint at the floor.

A remark on feasibility is in order. Since $dw_t = u_t dt$, the risky fraction is continuous in time, so no finite-speed feedback can guarantee the pathwise floor: if the state reaches $r = \alpha$ with $w > 0$, the floor is breached immediately with positive probability. The tracking speed derived below diverges as $r \downarrow \alpha$, which keeps violations small, but under quadratic costs the hard constraint should be read as holding up to small-cost corrections. Exact admissibility requires modifying the feedback near the floor so that $w \rightarrow 0$ before the floor is reached; the leading-order analysis below lives away from the floor and is unaffected.

In the frictionless case $\kappa = 0$, the bias function is explicit:

$$H^0(r) = L \log(r - \alpha) - \log r \quad (152)$$

up to an additive constant. It satisfies the reflecting condition, $H_r^0(1) = L/(1 - \alpha) - 1 = 0$, and the first-order condition of the frictionless Hamiltonian,

$$w = \frac{\mu}{\sigma^2} \frac{1 + rH_r^0}{1 - r^2H_{rr}^0} \quad (153)$$

reproduces the Grossman-Zhou target $y(r)$, since

$$1 + rH_r^0 = \frac{Lr}{r - \alpha} \quad 1 - r^2H_{rr}^0 = \frac{Lr^2}{(r - \alpha)^2} \quad (154)$$

These two expressions reappear below as the trading-cost multiplier and the Hamiltonian curvature.

The small-cost limit can be made explicit at leading order. It is a local tracking approximation around the frictionless target $y(r)$, in the same spirit as linear-quadratic execution models and small-cost price-impact expansions [6, 12]. Define

$$q := \frac{r - \alpha}{L} \quad (155)$$

Under the frictionless Grossman-Zhou policy, q has uniform long-run occupation density on $(0, 1)$. Away from the high-water boundary,

$$dR_t = R_t \mu y(R_t) dt + R_t \sigma y(R_t) dW_t \quad (156)$$

so the diffusion coefficient of the moving target $y(R_t)$ is

$$b_y(r) := y'(r) r \sigma y(r) = \frac{\theta^2 \alpha (r - \alpha)}{\sigma r^2} \quad (157)$$

Write $e_t = w_t - y(R_t)$. To second order, the loss from being off target is quadratic in e_t , and the cost of correcting the error is quadratic in u_t . The curvature of the frictionless Hamiltonian in w is

$$C(r) = \sigma^2 (1 - r^2 H_{rr}^0) = \sigma^2 \frac{Lr^2}{(r - \alpha)^2} \quad (158)$$

and, since the cost term enters the HJB as $-\frac{\kappa}{2} u^2 (1 + rH_r)$, the effective multiplier on the trading cost is

$$A(r) = 1 + rH_r^0 = \frac{Lr}{r - \alpha} \quad (159)$$

Thus the local problem is

$$\min_u \mathbb{E} \int \left[\frac{1}{2} C(r) e_t^2 + \frac{1}{2} \kappa A(r) u_t^2 \right] dt \quad de_t = u_t dt - b_y(r) dW_t \quad (160)$$

with r frozen over the short local time scale. The scalar LQ problem gives the leading tracking rule

$$dw_t = \lambda(R_t)(y(R_t) - w_t)dt \quad \lambda(r) = \sqrt{\frac{C(r)}{\kappa A(r)}} = \frac{\sigma}{\sqrt{\kappa}} \sqrt{\frac{r}{r - \alpha}} \quad (161)$$

The local tracking error is of order $\kappa^{1/4}$:

$$\text{Var}(w_t - y(R_t) \mid R_t = r) \sim \frac{b_y(r)^2}{2\lambda(r)} \quad (162)$$

At the optimum, the displacement loss and trading loss are equal to leading order, so the corresponding welfare loss is of order $\sqrt{\kappa}$:

$$g^* - g_\kappa \sim \frac{\sqrt{\kappa}}{2} \int_0^1 b_y(\alpha + Lq)^2 \sqrt{A(\alpha + Lq)C(\alpha + Lq)} dq \quad (163)$$

Since

$$\int_\alpha^1 \frac{(r - \alpha)^{1/2}}{r^{5/2}} dr = \frac{2L^{3/2}}{3\alpha} \quad (164)$$

we get

$$g_\kappa = g^* - \frac{\theta^4}{3\sigma} \alpha L^{3/2} \sqrt{\kappa} + o(\sqrt{\kappa}) \quad (165)$$

A useful by-product is the leading tracking-error magnitude:

$$\overline{(w - y)^2} \sim \frac{\theta^4}{7\sigma^3} \alpha L^{5/2} \sqrt{\kappa} \quad (166)$$

The coefficients used in the main-text Sharpe approximation are

$$B_g(L) = \frac{\theta^4}{3\sigma} \alpha L^{3/2} \quad (167)$$

$$B_v(L) = \frac{\theta^4}{7\sigma} \alpha L^{5/2} \quad (168)$$

15.6 Quadratic-cost Sharpe approximation

The quadratic-cost Sharpe plot uses the small-cost tracking approximation from Appendix 15.5. It is not the exact solution of the two-state HJB. The approximation freezes the frictionless occupation density of q and keeps the leading $\sqrt{\kappa}$ tracking corrections. Freezing the occupation density is innocuous for the growth rate g_κ , which is envelope-protected at order $\sqrt{\kappa}$, but not for $\langle s_\kappa^2 \rangle$: the perturbed policy also shifts the occupation measure of R_t at order $\sqrt{\kappa}$, and that shift is ignored here.

Write

$$a(q) := \frac{Lq}{\alpha + Lq} \quad q \sim \text{Unif}(0, 1) \quad (169)$$

and define

$$A_1(L) := \int_0^1 a(q) \, dq = 1 - \frac{\alpha}{L} \log\left(\frac{1}{\alpha}\right) \quad (170)$$

$$A_2(L) := \int_0^1 a(q)^2 \, dq = 2 - L - \frac{2\alpha}{L} \log\left(\frac{1}{\alpha}\right) \quad (171)$$

In the frictionless case, the local return volatility is $s(q) = \theta a(q)$, so

$$\langle s^2 \rangle = \theta^2 A_2(L) \quad (172)$$

and the arithmetic mean excess return is

$$\langle b \rangle = \theta^2 A_1(L) \quad (173)$$

With quadratic costs, the local-tracking approximation gives

$$g_\kappa \approx \frac{1}{2} \theta^2 L - B_g(L) \sqrt{\kappa} \quad (174)$$

$$\langle s_\kappa^2 \rangle \approx \theta^2 A_2(L) + B_v(L) \sqrt{\kappa} \quad (175)$$

where

$$B_g(L) := \frac{\theta^4}{3\sigma} \alpha L^{3/2} \quad (176)$$

$$B_v(L) := \frac{\theta^4}{7\sigma} \alpha L^{5/2} \quad (177)$$

The first coefficient is the growth loss derived in Appendix 15.5. The second is the contribution of the leading tracking-error variance:

$$\sigma^2 \overline{(w - y)^2} \sim \frac{\theta^4}{7\sigma} \alpha L^{5/2} \sqrt{\kappa} \quad (178)$$

Since log growth equals arithmetic mean minus one half of variance,

$$g_\kappa = m_\kappa - \frac{1}{2} \langle s_\kappa^2 \rangle \quad (179)$$

the cost-adjusted arithmetic mean is approximated by

$$m_\kappa \approx g_\kappa + \frac{1}{2} \langle s_\kappa^2 \rangle \quad (180)$$

$$\approx \theta^2 A_1(L) - \left[B_g(L) - \frac{1}{2} B_v(L) \right] \sqrt{\kappa} \quad (181)$$

Therefore the leading small-cost return Sharpe approximation is

$$\widetilde{\text{SR}}_{\text{ret},\kappa}(L) = \frac{\theta^2 A_1(L) - \left[B_g(L) - \frac{1}{2} B_v(L) \right] \sqrt{\kappa}}{\sqrt{\theta^2 A_2(L) + B_v(L) \sqrt{\kappa}}} \quad (182)$$

This is the formula plotted in Figure 4.

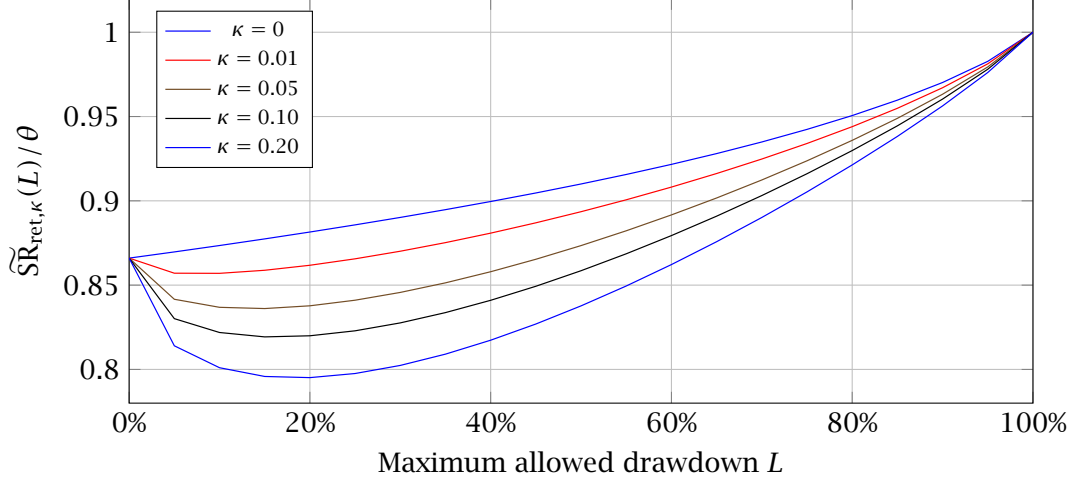


Figure 4: Leading-order continuous-time quadratic-cost approximation to the normalized ordinary return Sharpe ratio, using $\mu = 8\%$, $\sigma = 20\%$, and $\theta = 0.40$. The frictionless curve is $\kappa = 0$.

15.7 Sharpe calculations and the log-return Sharpe

This appendix gives the calculation behind the main-text return Sharpe formula and explains why the log-return Sharpe can move in the opposite direction. Consider any self-financing discounted wealth process whose local simple return is

$$\frac{dx_t}{x_t} = b_t dt + s_t dW_t \quad (183)$$

For long-run comparisons, write occupation averages as

$$\langle f \rangle := \lim_{T \rightarrow \infty} \frac{1}{T} \int_0^T f_t dt \quad (184)$$

The ordinary return Sharpe is the arithmetic-return mean divided by arithmetic-return volatility:

$$SR_{ret} := \frac{\langle b \rangle}{\sqrt{\langle s^2 \rangle}} \quad (185)$$

The log-return, or growth, Sharpe uses the log drift:

$$SR_{log} := \frac{\langle b - \frac{1}{2}s^2 \rangle}{\sqrt{\langle s^2 \rangle}} \quad (186)$$

Thus the two ratios are linked by the Ito correction

$$SR_{log} = SR_{ret} - \frac{1}{2}\sqrt{\langle s^2 \rangle} \quad (187)$$

The denominator is the same. The numerator is not. The log-return Sharpe penalizes variance directly through the term $\frac{1}{2}\langle s^2 \rangle$. A policy that reduces realized volatility can therefore improve SR_{log} even if its ordinary arithmetic-return Sharpe falls.

For the Grossman–Zhou policy, the local volatility coefficient is

$$s(q) = \theta a(q) \quad (188)$$

$$a(q) := \frac{Lq}{\alpha + Lq} \quad (189)$$

$$q \sim \text{Unif}(0, 1) \quad (190)$$

and the local arithmetic drift is

$$b(q) = \theta^2 a(q) \quad (191)$$

For $0 < L < 1$, define

$$A_1(L) := \int_0^1 a(q) \, dq = 1 - \frac{\alpha}{L} \log\left(\frac{1}{\alpha}\right) \quad (192)$$

$$A_2(L) := \int_0^1 a(q)^2 \, dq = 2 - L - \frac{2\alpha}{L} \log\left(\frac{1}{\alpha}\right) \quad (193)$$

Then

$$\frac{\text{SR}_{\text{ret}}(L)}{\theta} = \frac{A_1(L)}{\sqrt{A_2(L)}} \quad (194)$$

$$\frac{\text{SR}_{\text{log}}(L)}{\theta} = \frac{A_1(L) - \frac{1}{2}A_2(L)}{\sqrt{A_2(L)}} = \frac{L}{2\sqrt{A_2(L)}} \quad (195)$$

Equivalently, the ordinary return Sharpe is

$$\text{SR}_{\text{ret}}(L) = \theta \frac{1 - \frac{\alpha}{L} \log\left(\frac{1}{\alpha}\right)}{\sqrt{2 - L - \frac{2\alpha}{L} \log\left(\frac{1}{\alpha}\right)}} \quad (196)$$

These formulas explain why the ordinary return Sharpe and the log-return Sharpe move differently. As $L \downarrow 0$,

$$A_1(L) \sim \frac{L}{2} \quad A_2(L) \sim \frac{L^2}{3} \quad (197)$$

so both ratios approach

$$\frac{\sqrt{3}}{2} \quad (198)$$

But at $L = 1$, corresponding to unconstrained Kelly,

$$A_1(1) = A_2(1) = 1 \quad (199)$$

and therefore

$$\frac{\text{SR}_{\text{ret}}(1)}{\theta} = 1 \quad \frac{\text{SR}_{\text{log}}(1)}{\theta} = \frac{1}{2} \quad (200)$$

So, as the maximum allowed drawdown L increases from 0 to 1, the ordinary return Sharpe rises from $\sqrt{3}/2$ to 1, while the log-return Sharpe falls from $\sqrt{3}/2$ to $1/2$. Equivalently, tightening the drawdown constraint can raise the log-growth Sharpe while lowering the ordinary return Sharpe.

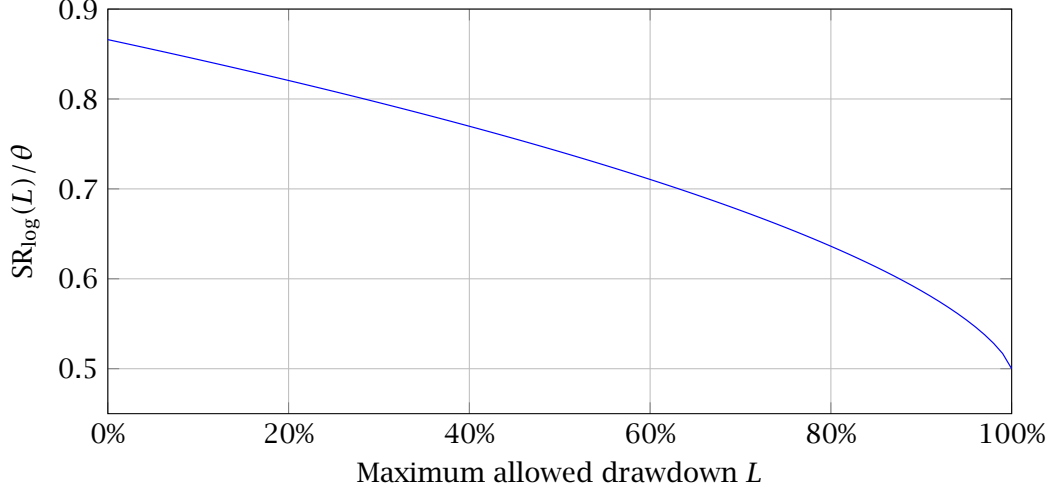


Figure 5: The normalized log-return Sharpe ratio $SR_{\log}(L)/\theta = g^*/(\theta\bar{\sigma}_{\log})$ of the optimal drawdown-constrained policy. It decreases from $\sqrt{3}/2$ at $L \rightarrow 0$ to $\frac{1}{2}$ at $L = 1$.

The monotonicity is unintuitive enough that it is worth checking by simulation. Table 4 simulates the auxiliary Kelly wealth and then applies the Grossman-Zhou transform pathwise. The reported “relative” number is

$$\frac{\hat{g}}{\theta \hat{\sigma}_{\log}}$$

where $\hat{g}$ is the annualized terminal log-growth estimate and $\hat{\sigma}_{\log}$ is the annualized realized log-volatility estimate. The simple Sharpe columns use arithmetic return increments instead: annualized mean simple excess return divided by annualized simple-return volatility. The run below uses $\theta = 0.40$, 10,000 paths, 100 years per path, and daily time steps.

Policy	L	Sim. g	Sim. $\sigma_{\log}$	Sim. relative	Formula relative	Sim. simple SR	Formula simple SR
Kelly	-	7.93%	40.00%	0.496	0.500	0.398	0.400
G-Z	5.00%	0.42%	1.23%	0.855	0.855	0.348	0.348
G-Z	10.00%	0.84%	2.49%	0.844	0.844	0.350	0.349
G-Z	20.00%	1.68%	5.11%	0.820	0.821	0.354	0.353
G-Z	50.00%	4.16%	14.05%	0.740	0.741	0.366	0.364

Table 4: Monte Carlo check of the relative log-growth Sharpe and the simple-return Sharpe. Kelly stays near 1/2 on the relative log-growth metric. The Grossman-Zhou relative log-growth ratios are larger because the drawdown rule spends more time at low instantaneous volatility than at high instantaneous volatility, while the ordinary simple Sharpe ratios remain below the Kelly simple Sharpe.

There is no contradiction. The ordinary Sharpe asks whether the average arithmetic excess return is high relative to return volatility. The log Sharpe asks whether the compounded growth rate is high relative to log-volatility. A drawdown constraint spends a large fraction of time with small exposure; this lowers ordinary return efficiency relative to Kelly, but it also lowers the variance drag in log growth. These are different orderings of the same policy family.

When is log-return Sharpe useful? The desirable feature of the log-return Sharpe is that its numerator is the actual long-run compounded growth rate. Log wealth is additive through time:

$$\log x_T - \log x_0 = \int_0^T \left(b_t - \frac{1}{2} s_t^2 \right) dt + \int_0^T s_t dW_t \quad (201)$$

so the numerator of $SR_{\log}$ is the ergodic drift of log wealth. This is not a cosmetic difference from the ordinary Sharpe. The correction $-\frac{1}{2}s_t^2$ is the variance drag from compounding. A strategy can have attractive arithmetic returns and still compound poorly if it spends too much time at high variance.

The metric is also path-policy aware. For a dynamic strategy, the quantities $\langle b \rangle$ and $\langle s^2 \rangle$ are occupation averages under the actual policy. This matters for drawdown rules, volatility targeting, stop-out rules, and other state-dependent strategies. Two policies can use the same underlying asset and have the same instantaneous opportunity set, but generate different long-run distributions of exposure. The log-return Sharpe sees this through the occupation measure.

A third useful property is that it exposes overbetting. If a base strategy is levered by a scalar $c > 0$, so that

$$\frac{dx_t^{(c)}}{x_t^{(c)}} = c b_t dt + c s_t dW_t \quad (202)$$

then

$$SR_{\text{ret}}(c) = \frac{\langle b \rangle}{\sqrt{\langle s^2 \rangle}} \quad (203)$$

is unchanged by leverage, while

$$SR_{\log}(c) = \frac{c \langle b \rangle - \frac{1}{2} c^2 \langle s^2 \rangle}{c \sqrt{\langle s^2 \rangle}} = SR_{\text{ret}} - \frac{1}{2} c \sqrt{\langle s^2 \rangle} \quad (204)$$

falls as leverage rises. The ordinary Sharpe is blind to the Kelly overbetting problem; the log-return Sharpe is not.

This last calculation also shows the danger. Log-return Sharpe should not be maximized as a free-standing objective when leverage or exposure can be scaled continuously. In the simple scaling example above, $SR_{\log}(c)$ is maximized by sending $c \downarrow 0$. That is not an investment policy; it is a nearly-flat position with a high ratio because the variance drag has been made tiny. If leverage is the free decision variable, the growth-optimal choice maximizes the numerator

$$c \langle b \rangle - \frac{1}{2} c^2 \langle s^2 \rangle \quad (205)$$

which gives

$$c^* = \frac{\langle b \rangle}{\langle s^2 \rangle} \quad (206)$$

not the ratio.

Thus $SR_{\log}$ is best used in three cases. First, use it as a reporting statistic for strategies whose scale is already fixed by mandate, margin, drawdown, risk, or portfolio-construction constraints. Second, maximize it only after imposing a nondegenerate risk budget, for example $\sqrt{\langle s^2 \rangle} = \bar{\sigma}$; with the denominator fixed, maximizing $SR_{\log}$ is the same as maximizing long-run growth. Third,

use it to compare policy shapes, not leverage scales: drawdown control versus Kelly, volatility targeting versus constant exposure, or different state-dependent allocation rules.

The practical rule is simple. If the investor's primitive objective is long-run wealth, maximize expected log growth g , subject to any hard constraints. If the investor wants a risk-adjusted diagnostic of how efficiently a fixed-scale policy compounds wealth, report $SR_{\log}$. If the investor cares about arithmetic mark-to-market performance over short horizons, funding constraints, or benchmark-relative returns, the ordinary return Sharpe may be the more relevant statistic. The two Sharpe ratios answer different questions, and a well-designed report should show both.

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